

Isolated MIDI input / 3.3 V UART / HAND ASSEMBLY

T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

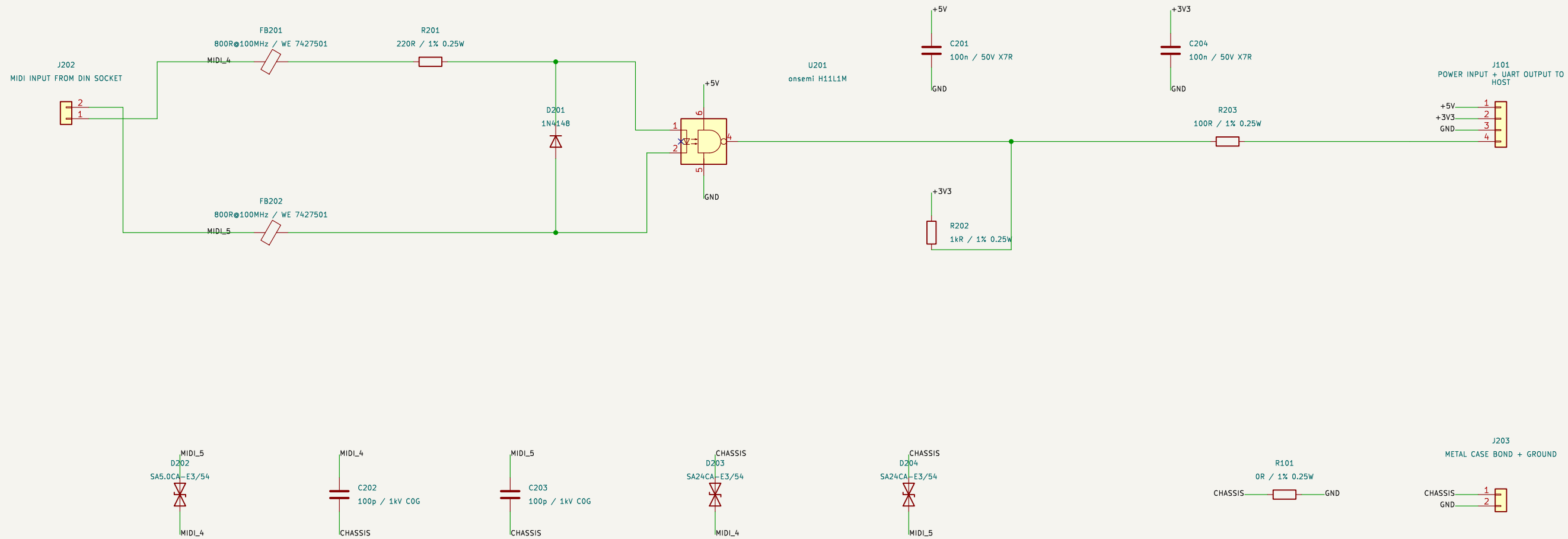
PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J202 / MIDI IN  
1: DIN4 IN    2: DIN5 IN

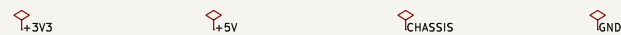
J203 / CASE BOND  
1: CASE 2: GND 0V

J101 / PWR IN / UART OUT  
1: 5V IN    2: 3V3 IN  
3: GND 0V   4: UART OUT

U201.3: NC – LEAVE OPEN



DIN panel socket: contact 4 -> J202.1; contact 5 -> J202.2. Pins 1/2/3 and shell unconnected.  
 Insulate DIN shell. J203.1 bonds the enclosure; R101 returns ESD to local GND.  
 Optocoupler LED current -> UART low. The output pull-up is 3.3 V, never 5 V.  
 Functional ground-loop isolation; transient clamps limit common-mode range. Not a safety isolator.



Engineering prototype – bench validation required

Ardor

Sheet: /  
File: Ardor\_MIDI\_THT.kicad\_sch

Title: Isolated MIDI input / 3.3 V UART / HAND ASSEMBLY

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| Size: A3 | Date: 2026-10-08 |
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KiCad E.D.A. 9.0.2

Rev: T1

1/1

Passive expression pedal / I2C ADC / HAND ASSEMBLY

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J302 / PASSIVE PEDAL  
1: TIP I/O 2: RING I/O  
3: GND 0V

JP301 / PEDAL JUMPER  
1: TIP 2: RING  
3: PEDAL IN 4: EXC OUT  
5: RING 6: TIP

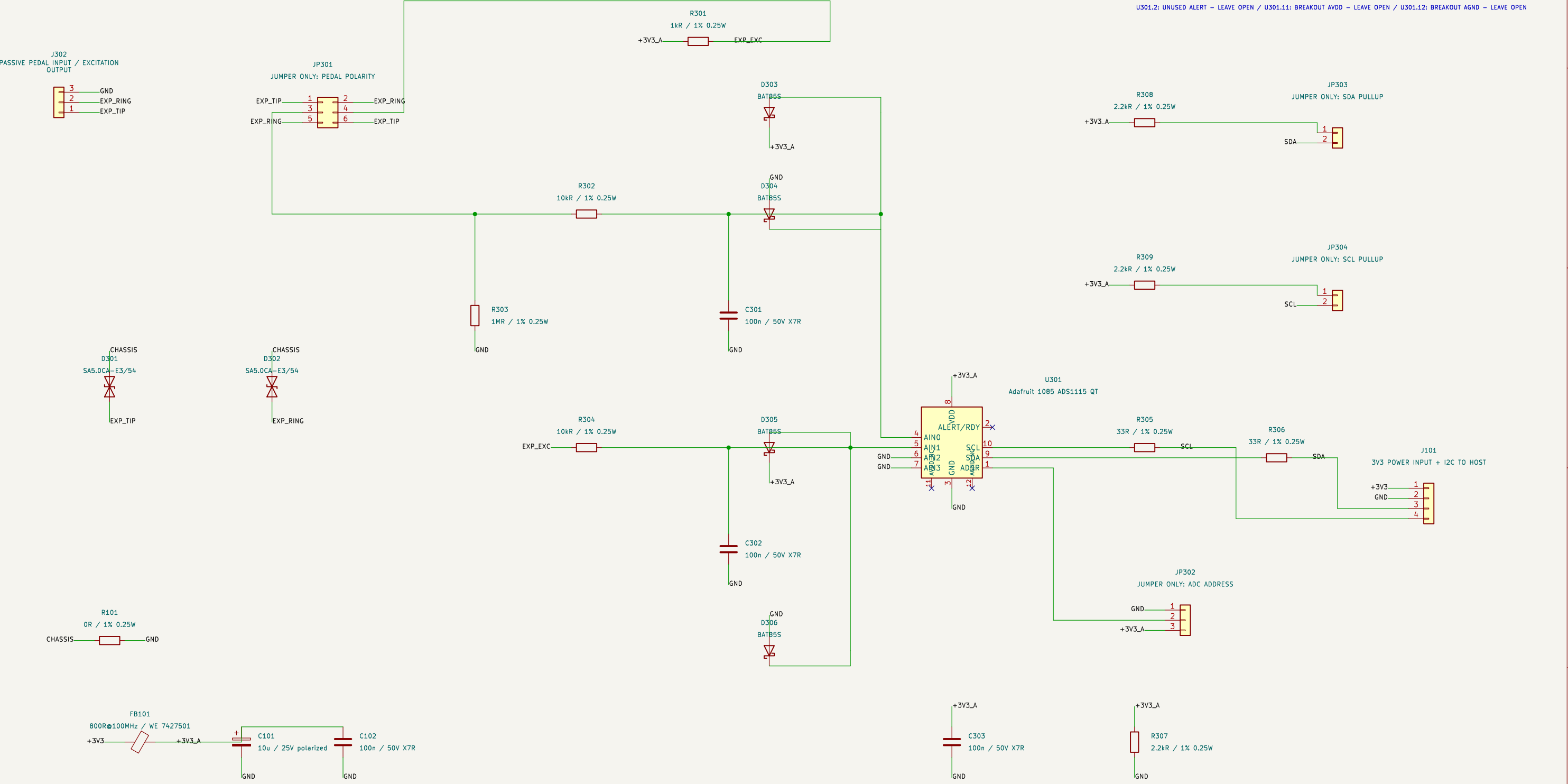
JP303 / DATA PULLUP  
1: PULLUP 2: DATA I/O

J101 / PWR IN / I2C  
1: 3V3 IN 2: GND 0V  
3: DATA I/O 4: CLOCK IN

JP302 / ADDR JUMPER  
1: LOW GND 2: SELECT  
3: HIGH 3V3

JP304 / CLOCK PULLUP  
1: PULLUP 2: CLOCK IN

U301.2: UNUSED ALERT - LEAVE OPEN / U301.11: BREAKOUT AVDD - LEAVE OPEN / U301.12: BREAKOUT AGND - LEAVE OPEN



Passive 10k–100k pot only. Sleeve = GND. JP301: both shunts 1–3 / 2–4 for tip wiper; 3–5 / 4–6 for ring wiper.  
JP302: 1–2 = 0x48; 2–3 = 0x49. Never leave ADDR floating. JP303/304 enable local 2.2k I2C pull-ups.  
Remove BOTH pull-up shunts when host already has pull-ups (Raspberry Pi). Start at 100 kHz I2C.  
ADS1115: +/-4.096 V, 128 SPS. Read AIN0 and AIN1; normalize AIN0/AIN1, calibrate endpoints and smooth.

Engineering prototype – bench validation required

Ardor

Sheet: /  
File: Ardor\_Expression\_THT.kicad\_sch

**Title: Passive expression pedal / I2C ADC / HAND ASSEMBLY**

Size: A3 Date: 2026–10–08

Rev: T1

KiCad E.D.A. 9.0.2

Id: 1/1

Stereo buffer / mono average / AC outputs / HAND ASSEMBLY

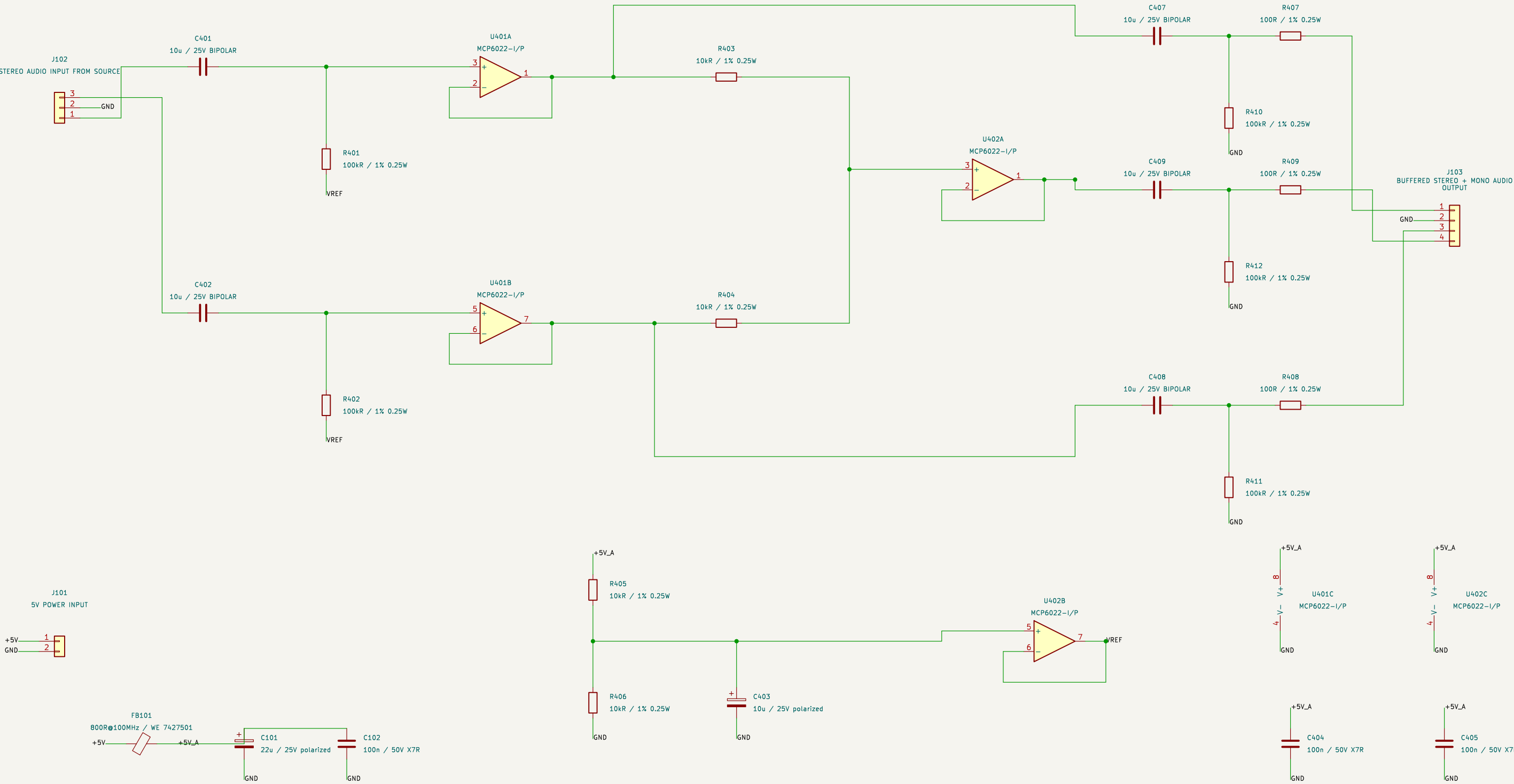
T1 | HAND ASSEMBLY | SEE PIN GUIDE AT RIGHT

PIN GUIDE: IN = Into this board / OUT = out / I/O = both / GND = 0 V

J101 / POWER IN  
1: 5V IN 2: GND 0V

J102 / AUDIO IN  
1: LEFT IN 2: GND 0V  
3: RIGHT IN

J103 / AUDIO OUT  
1: LEFT OUT 2: GND 0V  
3: RIGHT OUT 4: MONO OUT



Line-level input <=1 Vrms/channel. Output loads >=10k. Mono = (L + R)/2, unity buffers.  
All three outputs AC coupled; no 2.5 V bias leaves this board. For mono input, feed both channels  
or accept -6 dB when using one channel. Harness connectors are internal; add jack-entry ESD protection  
for exposed ports. This board is neither an instrument preamp nor an ADC, headphone or power amplifier.

Engineering prototype – bench validation required

Ardor

Sheet: /  
File: Ardor\_Mixer\_THT.kicad\_sch

**Title: Stereo buffer / mono average / AC outputs / HAND ASSEMBLY**

Size: A3 Date: 2026-10-08

Rev: T1

KiCad E.D.A. 9.0.2

Id: 1/1

Mono line output / relay mute / internal amp feed / HAND ASSEMBLY

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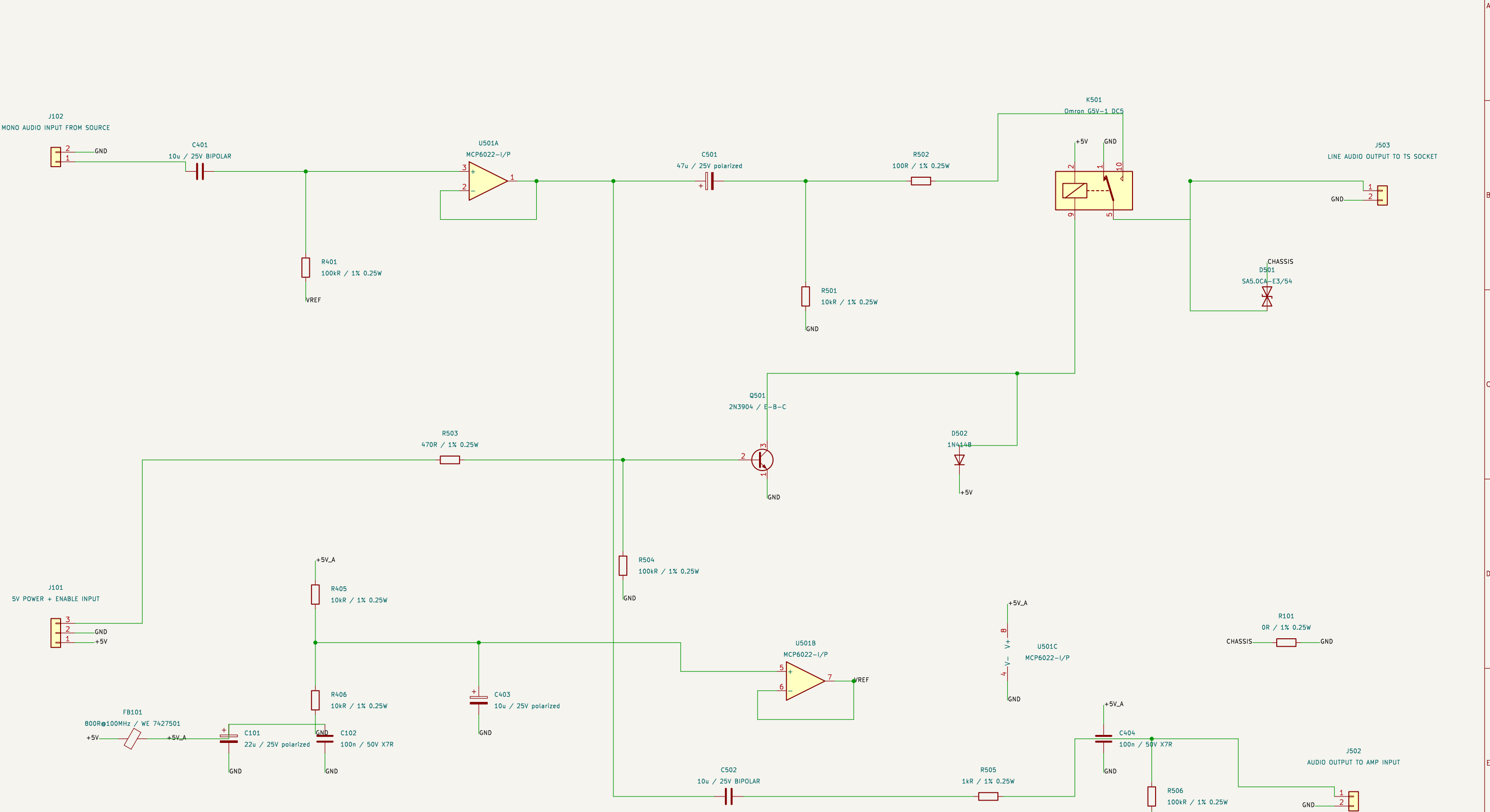
PIN GUIDE: IN = Into this board / OUT = out / I/O = both / GND = 0 V

J101 / PWR + ENABLE IN  
1: 5V IN 2: GND 0V  
3: ENABLE IN

J503 / LINE OUT  
1: LINE OUT 2: GND 0V

J102 / AUDIO IN  
1: AUDIO IN 2: GND 0V

J502 / AMP AUDIO OUT  
1: AUDIO OUT 2: GND 0V



Input <=1 Vrms, input impedance ~100k. LINE load >=10k; internal AMP load >=100k.  
Relay OFF: jack tip grounded, source disconnected. Relay ON: unity mono output. No true bypass.  
3.3 V active-high enable: leave LOW until audio is stable. 5 V supply budget >=50 mA.  
C501 + toward buffer. J502 is a signal feed, not a speaker driver; jack sleeves bond the enclosure.

+5V +5V\_A CHASSIS GND

Engineering prototype – bench validation required

Ardor

Sheet: /  
File: Ardor\_Line\_Out\_THT.kicad\_sch

Title: Mono line output / relay mute / internal amp feed / HAND ASSEMBLY

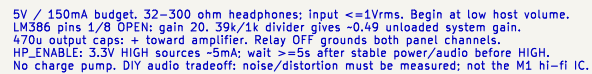
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| Size: A3           | Date: 2026-10-08 | Rev: T1 |
| KiCad E.D.A. 9.0.2 |                  | Id: 1/1 |

Stereo headphones / LM386 / relay mute

PIN GUIDE: IN = into this board / OUT = out / I/O = both / GND = 0 V

J102 / AUDIO IN  
1: LEFT IN    2: GND 0V  
3: RIGHT IN

U601.1: GAIN NC / U601.8: GAIN NC / U602.1: GAIN NC / U602.8: GAIN NC



Id: 1/1